

# Makrolon Hygard®

## Laminated polycarbonate sheet



### Your benefits:

- low inherent weight
- very good transparency
- anti-spall protection
- high abrasion resistance
- optimal UV protection



**Makrolon Hygard®** stands for transparent, multilayer laminated polycarbonate sheets. They meet all security requirements with regard to protection against forced entry and ballistic impact.

The **CG (containment glazing)** grades are particularly suitable for protection against forced entry with heavy tools such as sledge hammers and axes.

The **BR (bullet-resistant)** and the **MS (multiple shots)** grades were specially developed for ballistic protection and withstand even sustained fire from heavy automatic firearms.

### Fields of Application:

**Makrolon Hygard®** sheets offer maximum protection without forming dangerous slivers or "spider webs." They are ideally suited for use in all high-level security areas. The main fields of application are:

- banks
- jeweler's shops
- penal institutions
- court buildings
- museums
- security vehicles
- night desks
- and many others

| Product | Thickness | Weight                 | Light Transmission |
|---------|-----------|------------------------|--------------------|
| CG 375  | 10 mm     | 12 kg/m <sup>2</sup>   | 86%                |
| CG 500  | 13.5 mm   | 16.2 kg/m <sup>2</sup> | 83%                |
| BR 750  | 20 mm     | 24 kg/m <sup>2</sup>   | 88%                |
| MS 1250 | 33 mm     | 39.6 kg/m <sup>2</sup> | 75%                |

The sheets are available in the following sizes: 1,520 x 2,440 mm  
Other dimensions on request.

### Protection against forced entry

| Standard                            | Product          | Security Class                                         |
|-------------------------------------|------------------|--------------------------------------------------------|
| EN 356<br>Forced entry test         | CG 375<br>CG 500 | P7B<br>P8B                                             |
| ASTM F 1233.08<br>Forced entry test | CG 375<br>CG 500 | Class 1.4 Contraband Passage<br>Class 2.8 Body Passage |
| HPW TP 0500.02<br>Forced entry test | CG 375<br>CG 500 | Level I, Step 15                                       |

### Ballistic protection

| Standard                                        | Product           | Security Class                     |
|-------------------------------------------------|-------------------|------------------------------------|
| PM 2000 <sup>(*)</sup><br>Ballistic impact test | BR 750<br>MS 1250 | Level PM1-NS<br>Level PM4-NS & SG2 |
| UL 752<br>Ballistic impact test                 | BR 750<br>MS 1250 | Level I<br>Level VI                |

(\*) Because of the composition of this material (only thermoplastic and no glass) a classification according EN 1063 is not allowed, since EN 1063 only refers to glass/PVB combinations. The tests procedures according to PM 2007 are equivalent to EN 1063.

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Ideas, innovative, intelligent, interesting... Covestro i-line represents the next generation of quality products. This seal guarantees innovative and intelligent first-class solutions at all times for a multitude of requirements.

The protection against forced entry offered by **Makrolon Hygard®** is classed according to standards EN 356, ASTM F 1233 and HPW TP0500.02. During these tests, the sheets are exposed to physical and chemical attack. The tables below show the order in which such attack takes place (to HPM 1-54).

### Forced entry testing

| EN 356                                              |                |                 |                  |                 |                    |
|-----------------------------------------------------|----------------|-----------------|------------------|-----------------|--------------------|
| <b>Ball Test:</b> 10 cm steel ball (4.1 kg) 3 hits: | <b>P1</b>      | <b>P2A</b>      | <b>P3A</b>       | <b>P4A</b>      | <b>P5A</b>         |
| drop height (m)                                     | 1.5            | 3               | 6                | 9               | 9 (9hits)          |
| <b>Axe Test:</b>                                    | <b>P6B</b>     | <b>P7B</b>      | <b>P8B</b>       |                 |                    |
| number of hits                                      | >30            | >50             | >70              |                 |                    |
| <b>HPW TP 0500.02</b>                               | <b>Level I</b> | <b>Level II</b> | <b>Level III</b> | <b>Level IV</b> | <b>Level V</b>     |
| <b>Blunt tool impacting (impacts)</b>               |                |                 |                  |                 |                    |
| Sledge hammer/wedge (25)                            | 1, 4           | 8, 10           | 18, 24, 26       | 29, 32, 39      | 42, 45, 48, 51, 54 |
| 4" dia. pipe/sledge (25)                            | 2              | 7               | 17               | 28              | 41                 |
| Ram (10)                                            |                | 6               | 16               | 27              | 40                 |
| <b>Sharp tool impacting (impacts)</b>               |                |                 |                  |                 |                    |
| Chisel/hammer (25)                                  |                | 12              | 21, 23           | 33, 36, 38      | 47, 52             |
| Angle iron/sledge (25)                              |                | 13              | 22               |                 |                    |
| 1-1/2" dia. pipe/sledge (25)                        | 5              |                 |                  |                 |                    |
| Fire axe (25)                                       |                |                 |                  | 35              | 44, 50             |
| Wood maul (25)                                      |                | 15              | 20               | 31              | 46, 53             |
| <b>Thermal stress (minutes)</b>                     |                |                 |                  |                 |                    |
| Fire extinguisher, CO <sub>2</sub> (1)              | 3              | 9               |                  |                 |                    |
| Propane torch (5)                                   |                | 11              | 19               | 30              |                    |
| Acetylene (5)                                       |                |                 |                  |                 | 43                 |
| <b>Chemical deterioration (amount)</b>              |                |                 |                  |                 |                    |
| Gasoline (8 oz.)                                    |                | 14              |                  |                 |                    |
| Windshield washer (8 oz.)                           |                |                 | 25               | 34              |                    |
| Acetone (8 oz.)                                     |                |                 |                  | 37              | 49                 |
| Total forced entry sequences                        | 5              | 15              | 26               | 39              | 54                 |

### ballistic impact test

| UL 752           | Caliber        | Ammunition               | Weight in g. | min. velocity  | No. of shots |
|------------------|----------------|--------------------------|--------------|----------------|--------------|
| Level 1          | 9 mm           | full metal jacket        | 8.0          | 1.175 ft./sec. | 3            |
| Level 2          | .357 magnum    | jacketed lead soft point | 10.2         | 1.250 ft./sec. | 3            |
| Level 3          | .44 magnum     | lead semi-wadcutter      | 15.6         | 1.350 ft./sec. | 3            |
| Level 6          | 9 mm           | full metal jacket        | 8.0          | 1.400 ft./sec. | 5            |
| EN 1063/ PM 2000 | Caliber        | Ammunition               | Weight in g. | min. velocity  | No. of shots |
| BR1/ PM 1        | 22 long rifle  | lead round nose          | 2.6          | 360 m/s        | 3            |
| BR2/ PM 2        | 9 mm luger     | full steel jacket        | 8            | 400 m/s        | 3            |
| BR3/ PM 4        | 357 magnum     | full steel jacket        | 10.2         | 430 m/s        | 3            |
| BR4/ PM 4        | 44 rem. magnum | full copper jacket       | 15.6         | 440 m/s        | 3            |



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